

APPLIED ECONOMICS & QUANTITATIVE STATISTICS

New Egyptian Baccalaureate Educational Framework (EG-Bac Track) - Grade 12

ACADEMIC YEAR: 2025 / 2026 | EG-BAC GRADE 12 (ADVANCED SOCIAL SCIENCES & QUANTITATIVE REASONING)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Collegiate-level macroeconomic modelling and applied quantitative statistics tailored for the New Egyptian Baccalaureate, integrating econometric regression, multi-sector circular flows, Keynesian multipliers, monetary transmission mechanisms, and Gaussian continuous distribution calculus.

CORE CURRICULUM HIGHLIGHTS:

- Marginal rate of transformation and mathematical production frontier optimization
- Keynesian consumption and savings functions with investment multiplier calculus
- Laffer curve analysis, progressive taxation models, and fiscal debt sustainability
- Central bank quantitative easing, repo rates, and monetary transmission channels
- Least-squares linear regression slope, intercept, and error variance formulation
- Bayesian probability updating with tree-based sequential decision models
- Probability density integration, expected values, and cumulative distribution functions
- Standard Gaussian Z-score area calculations and confidence interval estimation

APPROVED CHAPTER MODULES:

- Unit 1: Scarcity, Economic Needs & Methodology	pp. 1-26 (14 core units)
- Unit 2: Factors of Production & Capital Accumulation	pp. 27-52 (14 core units)
- Unit 3: National Income, Product & Macroeconomic Equilibrium	pp. 53-78 (15 core units)
- Unit 4: Public Finance, Taxation & Fiscal Policy	pp. 79-104 (14 core units)
- Unit 5: Money, Modern Banking & Monetary Systems	pp. 105-130 (14 core units)
- Unit 6: International Economic Relations & Global Trade	pp. 131-156 (15 core units)
- Unit 7: Linear Correlation & Least-Squares Regression	pp. 157-182 (14 core units)
- Unit 8: Conditional Probability, Bayes Theorem & Independence	pp. 183-208 (14 core units)
- Unit 9: Random Variables & Probability Distributions	pp. 209-234 (14 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: EG-Bac Advanced Quantitative Macroeconomics & Applied Statistics Compendium
Full Ministerial Textbook: 260 Pages - 29.5 MB (Direct on MOE Portal)
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Official Portal: <https://moe.gov.eg/ar/elearning/egbac/economics/> - Code: EGBAC-ECON-S
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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Collegiate-level macroeconomic modelling and applied quantitative statistics tailored for the New Egyptian Baccalaureate, integrating econometric regression, multi-sector circular flows, Keynesian multipliers, monetary transmission mechanisms, and Gaussian continuous distribution calculus. Marginal rate of transformation and mathematical production frontier optimization Keynesian consumption and savings functions with investment multiplier calculus Laffer curve analysis, progressive taxation models, and fiscal debt sustainability Central bank quantitative easing, repo rates, and monetary transmission channels Least-squares linear regression slope, intercept, and error variance formulation Bayesian probability updating with tree-based sequential decision models Probability density integration, expected values, and cumulative distribution functions Standard Gaussian Z-score area calculations and confidence interval estimation

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Unit 1: Scarcity, Economic Needs & Methodology	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 2: Factors of Production & Capital Accumulation	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 3: National Income, Product & Macroeconomic Equilibrium	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 4: Public Finance, Taxation & Fiscal Policy	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 5: Money, Modern Banking & Monetary Systems	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 6: International Economic Relations & Global Trade	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 7: Linear Correlation & Least-Squares Regression	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 8: Conditional Probability, Bayes Theorem & Independence	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 9: Random Variables & Probability Distributions	3 wks	6 pts	10.0%	14 core instructional units covering national standards and
Unit 10: Gaussian Normal Curve & Standard Z-Transformations	3 wks	6 pts	10.0%	15 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: UNIT 1: SCARCITY, ECONOMIC NEEDS & METHODOLOGY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 1: Scarcity, Economic Needs & Methodology.
- [+] Synthesize and apply analytical frameworks across all 14 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Unit 1: Scarcity, Economic Needs & Methodology.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: UNIT 2: FACTORS OF PRODUCTION & CAPITAL ACCUMULATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 2: Factors of Production & Capital Accumulation.
- [+] Synthesize and apply analytical frameworks across all 14 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Unit 2: Factors of Production & Capital Accumulation.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: UNIT 3: NATIONAL INCOME, PRODUCT & MACROECONOMIC EQUILIBRIUM

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 3: National Income, Product & Macroeconomic Equilibrium.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Unit 3: National Income, Product & Macroeconomic Equilibrium.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\left(\frac{1}{N}\right) \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

APPLIED ECONOMICS & QUANTITATIVE STATISTICS - CORE PRINCIPLES
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio: $\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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